

Assignment_3

ALL files for this assignment can be found at https://github.com/Dylan-Hamilton3412/64060_-Dhamil28

Load required libraries for data manipulation, modeling, and evaluation
`library(dplyr)`

`##`

`## Attaching package: 'dplyr'`

`## The following objects are masked from 'package:stats':`

`##`

`## filter, lag`

`## The following objects are masked from 'package:base':`

`##`

`## intersect, setdiff, setequal, union`

`library(caret)`

`## Warning: package 'caret' was built under R version 4.5.2`

`## Loading required package: ggplot2`

`## Warning: package 'ggplot2' was built under R version 4.5.2`

`## Loading required package: lattice`

`library(readr)`

`library(class)`

`library(gmodels)`

`## Warning: package 'gmodels' was built under R version 4.5.3`

`library(e1071)`

`## Warning: package 'e1071' was built under R version 4.5.3`

`##`

`## Attaching package: 'e1071'`

`## The following object is masked from 'package:ggplot2':`

`##`

`## element`

Load dataset

`BankData <- read.csv("UniversalBank.csv")`

Select only relevant variables:

Column 10 = Online

Column 13 = CreditCard

Column 14 = Personal.Loan (target variable)

```

BankData <- select(BankData, 10, 13, 14)

## Convert variables to factors for classification
BankData$Personal.Loan <- as.factor(BankData$Personal.Loan)
BankData$Online <- as.factor(BankData$Online)
BankData$CreditCard <- as.factor(BankData$CreditCard)

## Set seed for reproducibility
set.seed(123)

## Create training (60%) and validation (40%) split
Index_Train <- createDataPartition(BankData$Personal.Loan, p = .6, list = FALSE)
Train_Data <- BankData[Index_Train,]
Val_Data <- BankData[-Index_Train,]

## Part A:
## Create a pivot tables with Online as the Y and CreditCard and Personal.Loan as the X
Pivot_Table <- table(CC = Train_Data$CreditCard,
                     Loan = Train_Data$Personal.Loan,
                     Online = Train_Data$Online
                     )
## Display pivot table (counts of all combinations)
Pivot_Table

## , , Online = 0
##
##      Loan
## CC      0      1
## 0  791   79
## 1  310   33
##
## , , Online = 1
##
##      Loan
## CC      0      1
## 0 1144  125
## 1  467   51

## Part B:

## Use Naive Bayes formula to calculate the probability of Loan acceptance
Prob_Loan <- Pivot_Table["1","1","1"] / (Pivot_Table["1","1","1"] + Pivot_Table["1","0","1"])
Prob_Loan <- Prob_Loan *100
Prob_Loan

## [1] 9.84556

```

##Part C:

Generate Pivot tables for the the two non target variables vs the target variable

Loan vs Online

```
Online_Pivot <- table(Loan = Train_Data$Personal.Loan,  
                      Online = Train_Data$Online  
                      )
```

Loan vs CreditCar

```
CC_Pivot <- table(Loan = Train_Data$Personal.Loan,  
                 CC = Train_Data$CreditCard  
                 )
```

Display tables

Online_Pivot

```
##      Online  
## Loan    0    1  
##    0 1101 1611  
##    1  112  176
```

CC_Pivot

```
##      CC  
## Loan    0    1  
##    0 1935  777  
##    1  204   84
```

Part D:

P(CC=1 | Loan=1)

```
P1 <- CC_Pivot["1", "1"] / (CC_Pivot["1", "1"] + CC_Pivot["1", "0"])  
P1 <- P1 * 100  
P1
```

```
## [1] 29.16667
```

P(Online=1 | Loan=1)

```
P2 <- Online_Pivot["1", "1"] / (Online_Pivot["1", "1"] + Online_Pivot["1", "0"]  
")  
P2 <- P2 * 100  
P2
```

```
## [1] 61.11111
```

P(Loan=1)

```
P3 <- sum(Train_Data$Personal.Loan == "1") / nrow(Train_Data)  
P3 <- P3 * 100  
P3
```

```
## [1] 9.6

## P(CC=1 | Loan=0)
P4 <- CC_Pivot["0", "1"] / (CC_Pivot["0", "1"] + CC_Pivot["0", "0"])
P4 <- P4 * 100
P4

## [1] 28.65044

## P(Online=1 | Loan=0)
P5 <- Online_Pivot["0", "1"] / (Online_Pivot["0", "1"] + Online_Pivot["0", "0"])
P5 <- P5 * 100
P5

## [1] 59.40265

## P(Loan=0)
P6 <- sum(Train_Data$Personal.Loan == "0") / nrow(Train_Data)
P6 <- P6 * 100
P6

## [1] 90.4

## Part E
##Manually compute the Naive Bayes probability using variables from Part D
Pivot_Manual_NB_Calc <- (P1 * P2 * P3) / ((P1 * P2 * P3) + (P4 * P5 * P6))

# Convert to percentage
Pivot_Manual_NB_Calc <- Pivot_Manual_NB_Calc * 100
Pivot_Manual_NB_Calc

## [1] 10.00861

## Part F
```

The probability obtained from the pivot table in Part B is 9.85%, while the probability obtained using the Naive Bayes method in Part E is approximately 10.01%. The pivot table value is a more accurate estimate because it is calculated directly from the observed frequencies of customers who have a credit card and actively use online banking. The Naive Bayes estimate assumes that Credit Card ownership and Online banking usage are conditionally independent given the loan outcome, which may not hold exactly in the data. Therefore, the probability obtained from the pivot table is considered the more accurate estimate.

```
##Part G

#Fit Naive Bayes model
Loan_NB_Model <- naiveBayes(Personal.Loan ~ Online + CreditCard, data = Train_Data)
Loan_NB_Model
```

```
##
## Naive Bayes Classifier for Discrete Predictors
##
## Call:
## naiveBayes.default(x = X, y = Y, laplace = laplace)
##
## A-priori probabilities:
## Y
##      0      1
## 0.904 0.096
##
## Conditional probabilities:
##      Online
## Y      0      1
## 0 0.4059735 0.5940265
## 1 0.3888889 0.6111111
##
##      CreditCard
## Y      0      1
## 0 0.7134956 0.2865044
## 1 0.7083333 0.2916667
```

Since the Naive Bayes classification calculation requires the product of the conditional probabilities of the predictors—in this case $P(CC=1|Loan=1)$, $P(Online=1|Loan=1)$, and the prior probability $P(Loan=1)$, divided by the total probability across both classes (loan acceptance and non-acceptance) you would essentially need every variable calculated from Part D of this assignment. It uses both the likelihood of the predictors within each class and the overall class distribution to compute the final posterior probability $P(Loan=1|CC=1,Online=1)$.

```
## Calculate Probability of  $P(Loan = 1 \mid CC = 1, Online = 1)$  given the Loan_NB_Model values
NBVal_Manual_Calc <- (0.2916667 * 0.6111111 * 0.096) / ((0.2916667 * 0.6111111 * 0.096) + (0.5940265 * 0.2865044 * 0.904))
NBVal_Manual_Calc
## [1] 0.1000861
```

The probability calculated using the Naive Bayes classification model closely matches the value obtained by manually computing the probabilities in Parts D and E. The small difference between the two results is due to rounding, but overall both methods produce nearly identical estimates for $P(Loan=1|CC=1,Online=1)$.

```
##Using Loan_NB_Model amounts to calculate probability
Test_df <- data.frame(Online = "1", CreditCard = "1")
Final_Predict <- predict(Loan_NB_Model, Test_df)
Final_Predict
## [1] 0
## Levels: 0 1
```

The final prediction shows that if predicted this calls would result in a 0 for personal loan meaning they would be classified as not getting a personal loan

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